CS341 · COMPUTER GRAPHICS

Midterm 461 Solutions

20 marks

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01 Solution Q1

1. Define computer graphics

Computer graphics is commonly understood to mean the **creation, storage, and manipulation of models and images**.

2. Absolute vs relative coordinate input

Absolute coordinate input: each point is plotted relative to the origin; the values are the actual positions in the coordinate system.
Relative coordinate input: each point is plotted relative to the last input (the current position), as an offset.
Example: current position (3, 8), relative (2, −1) → absolute (5, 7).

3. Aspect ratio

$$AR = \frac{\frac{Yp}{Ymax}}{\frac{Xp}{Xmax}}$$

$$= \frac{\frac{1080}{5}}{\frac{1920}{10}}$$

$$= \frac{216}{192}$$

$$= 1.125$$

02 Solution Q2: true / false

#	Answer	Reason
1	**False**	Plasma is emissive and must ionize gas to glow, so it uses more power than LCD.
2	**False**	In subtractive mixing, M + Y + C = black.
3	**False**	It scans from top to bottom, one horizontal line at a time.
4	**False**	Random scan draws lines directly at very high resolution; raster gives stair-stepped lines.
5	**True**	Refresh rate = how many times per second the picture is redrawn.
6	**False**	There are also CAD, medical visualization, education, simulation and more.

03 Solution Q3

#	Answer
1	**Cathode Ray Tube (CRT)**
2	**CMY / CMYK**
3	**Refresh rate**
4	**Synthetic images indistinguishable from reality**
5	**Electrons**
6	**All of the above**
7	**RGB**

04 Solution Q4: fill in the blanks

#	Answer	Why
1	**Red**	Subtractive: yellow absorbs blue and magenta absorbs green, leaving red.
2	**Blue**	Magenta absorbs green and cyan absorbs red, leaving blue.
3	**Cyan**	Additive: `(0,1,0) + (0,0,1) = (0,1,1)`.
4	**Resolution**	
5	**Persistence**	
6	**Color gamut**	
7	**Tint**	
8	**Tone**	

05 Solution Q5

Hard-copy devices: laser printer, ink-jet printer, film recorder, electrostatic printer, pen plotter.

Applications: computer aided design (CAD), medical visualization, entertainment (games and cinema), simulation, education, training, scientific visualization, user interfaces.

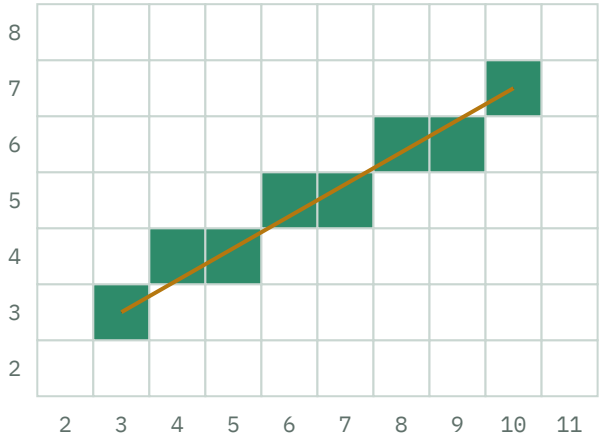
RGB = Red, Green, Blue.

06 Solution Q6: Bresenham

$$\Delta x = 7, \quad \Delta y = 4, \quad m = \frac{4}{7} < 1$$

$$2\Delta y = 8, \quad 2\Delta y - 2\Delta x = -6, \quad p0 = 2\Delta y - \Delta x = 1$$

k	p_k	Decision	Next pixel	p_{k+1}
0	1	$p_k \geq 0$	(4, 4)	-5
1	-5	$p_k < 0$	(5, 4)	3
2	3	$p_k \geq 0$	(6, 5)	-3
3	-3	$p_k < 0$	(7, 5)	5
4	5	$p_k \geq 0$	(8, 6)	-1
5	-1	$p_k < 0$	(9, 6)	7
6	7	$p_k \geq 0$	(10, 7)	1



Pixels: (3,3) (4,4) (5,4) (6,5) (7,5) (8,6) (9,6) (10,7)